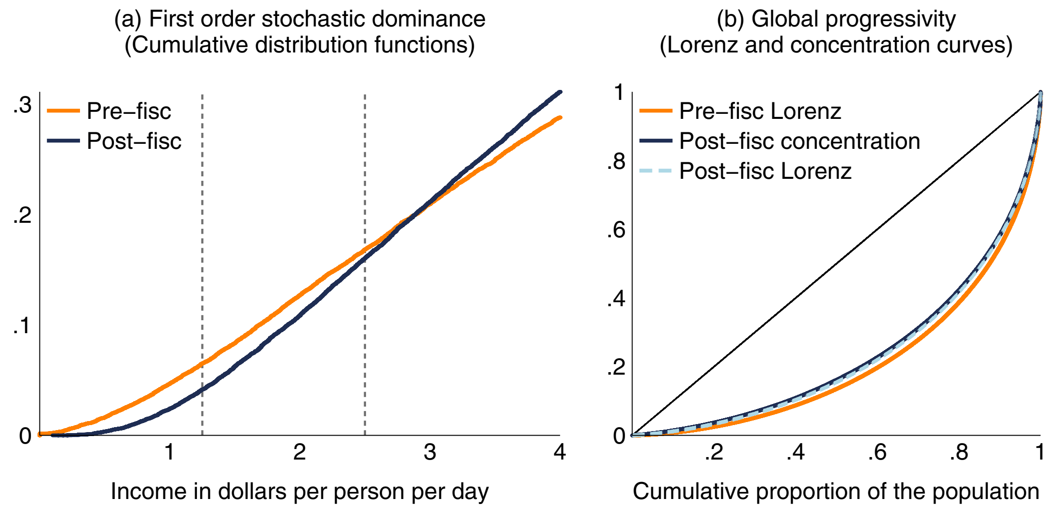

Fig. 2. Conventional tools to assess the tax and transfer system in Brazil. Note: Dashed vertical lines included at common "international" poverty lines of $1.25 and $2.50 per person per day.

that FI occurred using conventional measures and a poverty line above $3 per day, but would still be unaware of its extent without FI measures.[18]

Using the $2.50 line, we know that 5.6% of Brazil's population and over one-third of its post-fisc poor experience FI (Table 2); these impoverished individuals pay a total of $676 million more in taxes than they receive in transfers annually (Table 3), which is equivalent to 10% of the 2009 budget of Bolsa Família, Brazil's flagship anti-poverty program that reaches over one-fourth of the country's population. While substantial in size, this FI is dwarfed by FGP from Brazil's transfer programs, which totals over $3.5 billion. The absolute poverty gap, or the minimum amount that would need to be transferred to the poor to eliminate poverty if transfers were perfectly targeted, falls from $12.4 billion before taxes and transfers to $9.6 billion after. The change in the absolute poverty gap, $2.8 billion, looks impressive, but masks differential trends in two groups of the poor: those who gain (a total of $3.5 billion) and those who lose (a total of $676 million), as revealed by the decomposition of the change in the poverty gap derived in Section 4.

Fig. 3 shows how this decomposition and our axiomatic measures of total FI and FGP in Brazil vary with the poverty line. For low poverty lines, FI is essentially non-existent: at $1.25 per day, for example, total FI is $28 million per year, or 0.4% of the 2009 budget of Bolsa Família (Fig. 3a). This is not surprising in light of the unconditional component of the government cash transfer program Bolsa Família, available to households with income below 70 reais per person per month ($1.22 per day), regardless of whether the household has children or elderly members, and without conditions. At higher poverty lines, FI begins to increase more rapidly, and at a poverty line of $2.88 the rate of increase of FI exceeds the rate of increase of FGP: this can be seen by comparing the slopes of the solid curves in Fig. 3a, or by looking at the point where the difference between the two curves (plotted as the dashed curve in Fig. 3a) is at its maximum. By Proposition 4, this is also the point at which the absolute poverty gap reduction acheived by the fiscal system reaches its maximum, as seen by the dashed curve in Fig. 3b.

At this poverty line of $2.88 per day, where maximum poverty reduction is achieved, the difference between the pre-fisc and post-fisc poverty gaps is $2.9 billion. The eligibility cut-off for the conditional component of Bolsa Família, available to families with children who comply with certain education and health requirements, is $2.45 per person per day. Just above this line, a number of families still receive benefits due to program leakages, variable and mismeasured income, or components of income we are measuring that are not taken into account in the estimation of eligible income; not far above the line, however, families become much less likely to receive the program and we see a simultaneous deceleration of fiscal gains and acceleration of impoverishment.

[18] It is easy to show that if the post-fisc distribution does not first order stochastically dominate the pre-fisc distribution on the domain from 0 to the maximum poverty line, then FI has occurred.

6. Conclusions

Anti-poverty policies are increasingly being discussed in the same breath as the taxes used to pay for them. One example is the focus on mobilizing domestic resources to finance the policies necessary to achieve the United Nations' Post-2015 Sustainable Development Goals. To analyze transfers, subsidies, and taxes together, poverty comparisons and progressivity measures are often used. These measures, however, can lead us to conclude that the tax and transfer system unambiguously benefits the poor, when in fact a substantial number of poor are not compensated with transfers for their tax burdens. Indeed, we observe this in a number of developing countries: out of seventeen developing countries for which we have data, fifteen have tax and transfer systems that unambiguously reduce poverty and are globally progressive, but in ten of these at least one-quarter of the poor pay more in taxes than they receive in transfers and subsidies. In Brazil, for example, over one-third of the post-fisc poor experience fiscal impoverishment, paying a total of $676 million more in taxes than they receive in transfers and subsidies.

Given this shortcoming of conventional criteria and the debate about anti-poverty policies and the taxes used to pay for them, we propose a set of axioms that should be met by a measure of FI, and show that these uniquely determine the measure up to a proportional transformation. We also propose a partial ordering to determine when one fiscal system, such as that under a proposed reform, induces unambiguously less FI than another, such as the current system, over a range of possible poverty lines. To obtain a complete picture of the fiscal system's effect on the poor, we propose an analogous measure of fiscal gains of the poor, and show that the difference between the pre-fisc and post-fisc poverty gaps can be decomposed into our axiomatic measures of FI and FGP.

Our results can be extended to comparisons between two points in time or before and after a policy reform, rather than pre- and